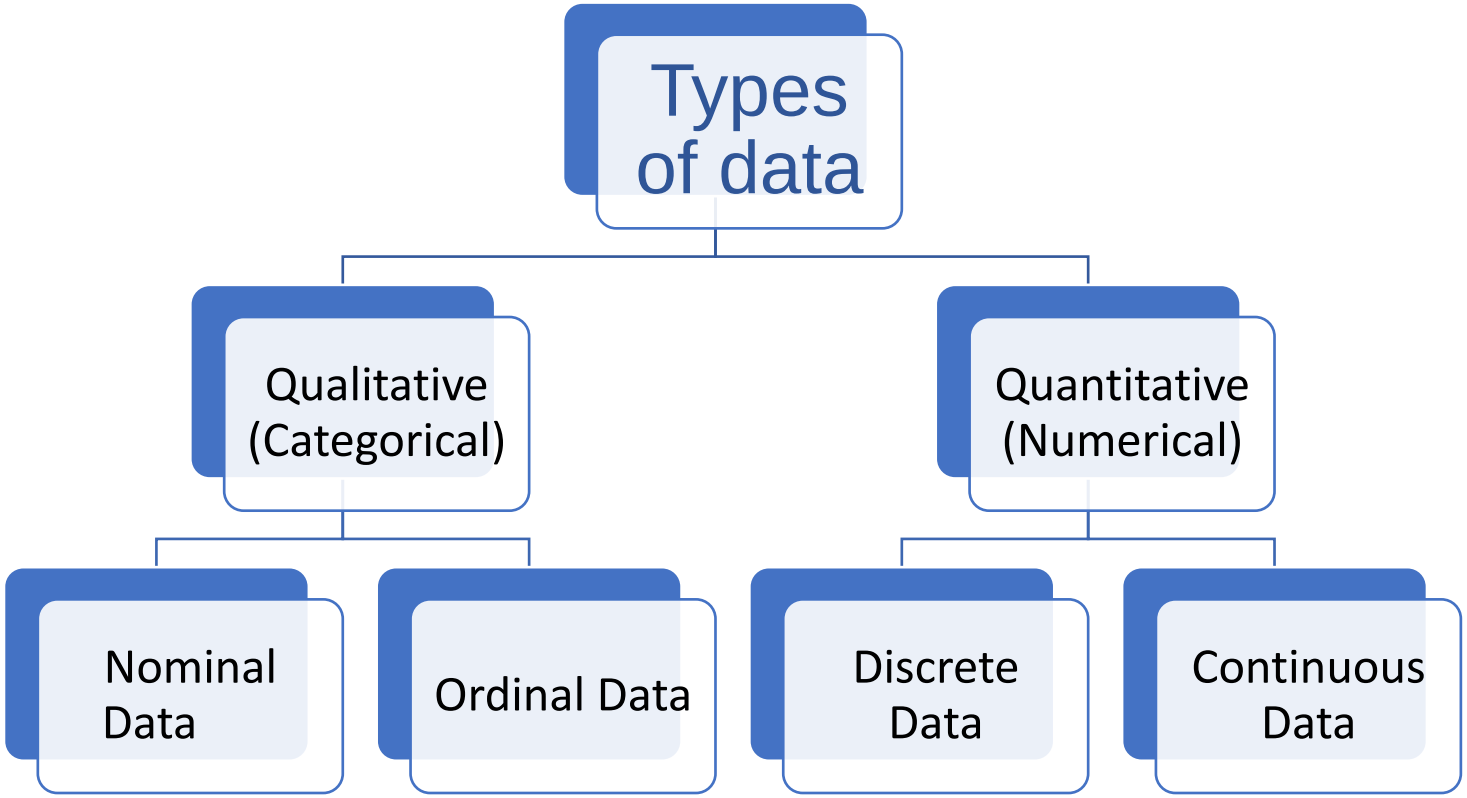
Visualizing Different Combinations of Numerical and Categorical Data



Nominal Data: Divides variable into mutually exclusive, Labeled category. Eg: Eye color, Transport etc

Ordinal Data: classified variables into categories of natural order. Eg: grades, rank, educational level etc

Discrete Data: Discrete data is information that is collected and can be counted. This type of data only has a certain number of values. Typically, it involves integers—whole numbers, rather than fractions.

Continuous data: Continuous data is data that can take any value and can be infinitely broken down into smaller parts. It differs from discrete data in that it can change its value. Continuous data can also fluctuate over time.

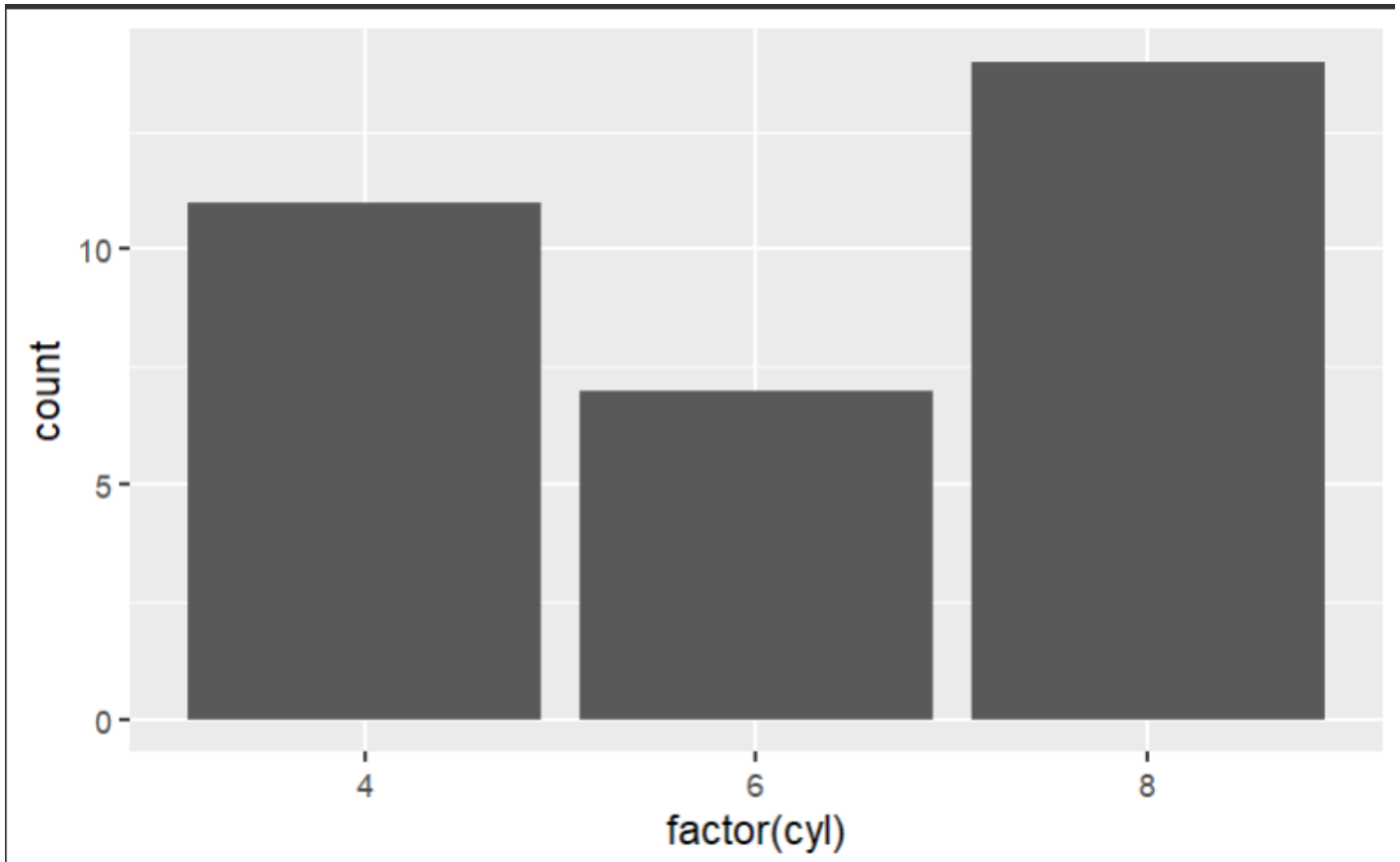
Combinations

1. Categorical
2. Numerical
3. Numerical-Numerical
4. Categorical-Numerical
5. Categorical-Categorical

Categorical (univariate)

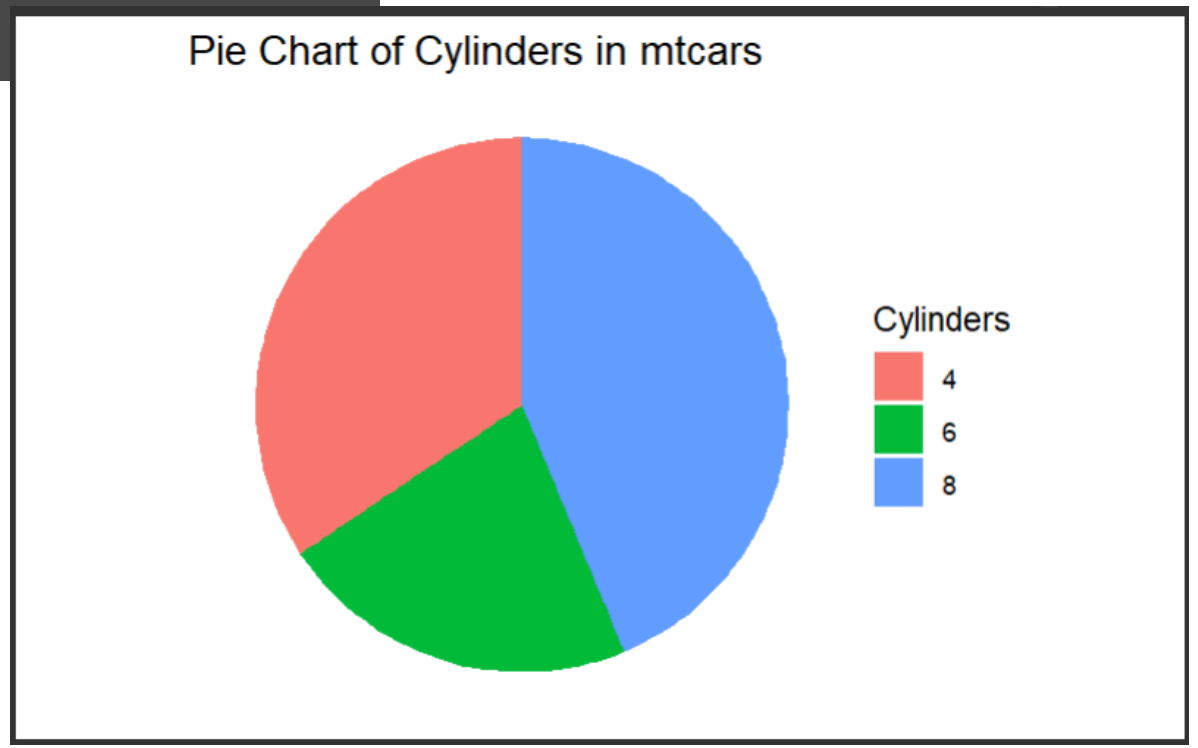
Bar Plot

```
ggplot(data = mtcars, aes(x = factor(cyl))) +  geom_bar()
```



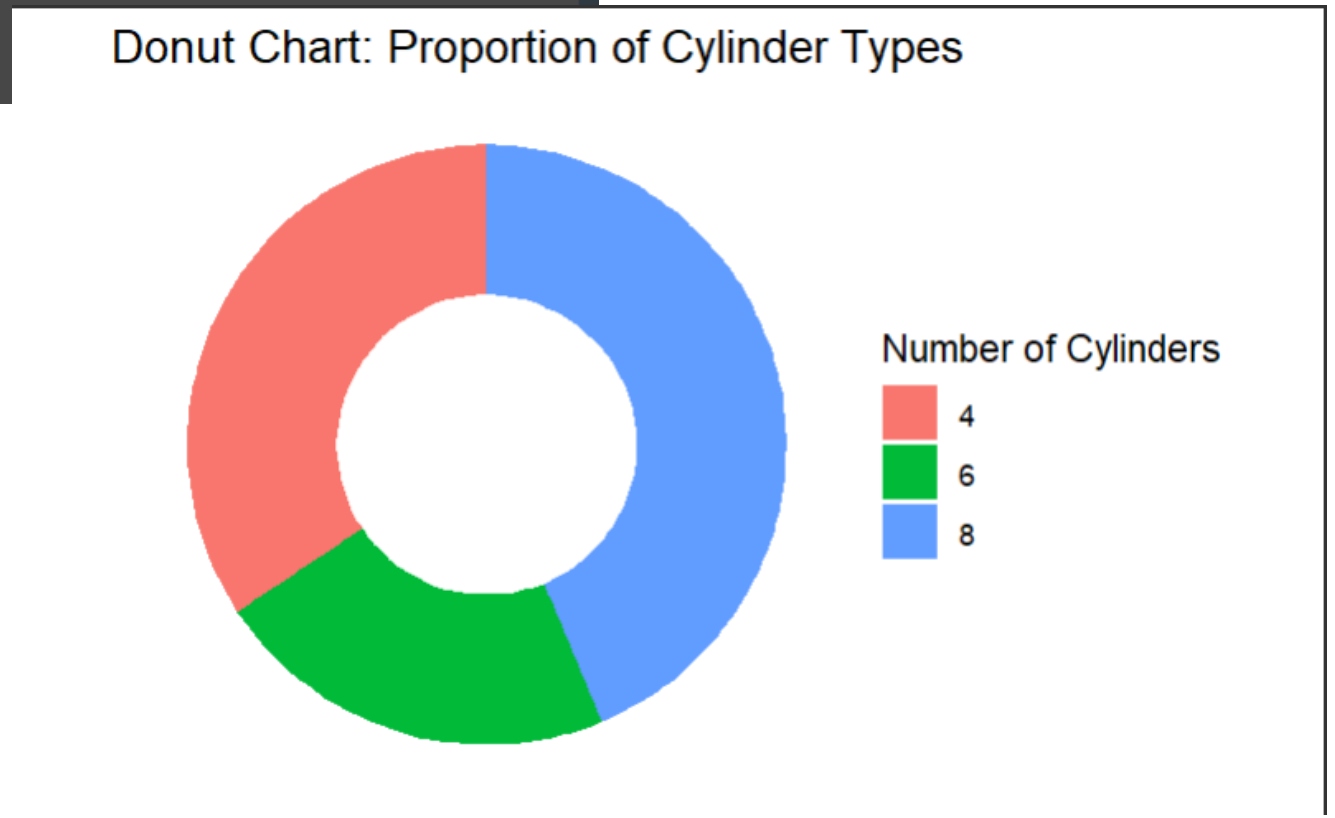
Pie Chart

```r
ggplot(data = mtcars, aes(x = "", fill = factor(cyl))) +
  geom_bar(width = 1) +
  coord_polar(theta = "y") +
  labs(title = "Pie Chart of Cylinders in mtcars", fill =
"Cylinders") +
  theme_void() # Removes unnecessary axis and grid lines
for a clean pie chart look
# coord_polar(theta = "y") converts the bar chart into a
circular pie chart.
```



Donut Chart

```
ggplot(mtcars, aes(x = 2, fill = factor(cyl))) +
  geom_bar(width = 1, position = "fill") +
  coord_polar(theta = "y") +
  xlim(0.5, 2.5) +
  labs(title = "Donut Chart: Proportion of Cylinder Types",
       fill = "Number of Cylinders") +
  theme_void()
```

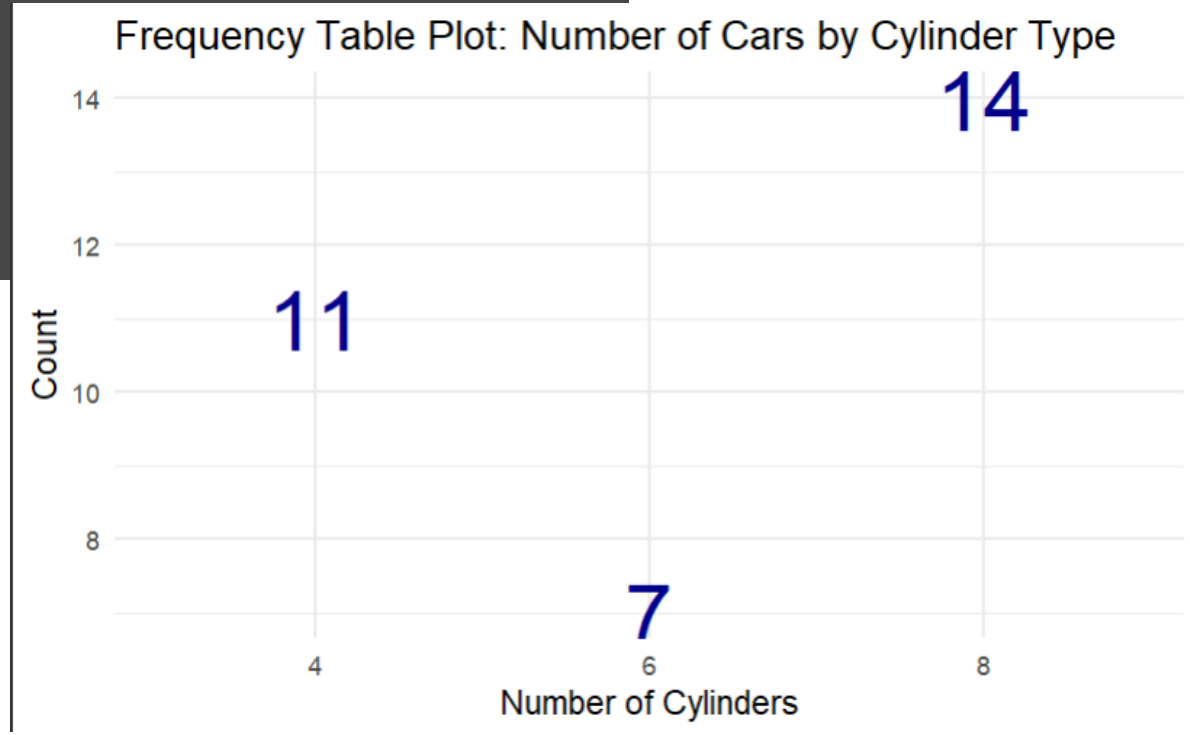


Frequency Plot

```r
library(ggplot2)
library(dplyr)

freq_table <- mtcars %>% count(cyl)

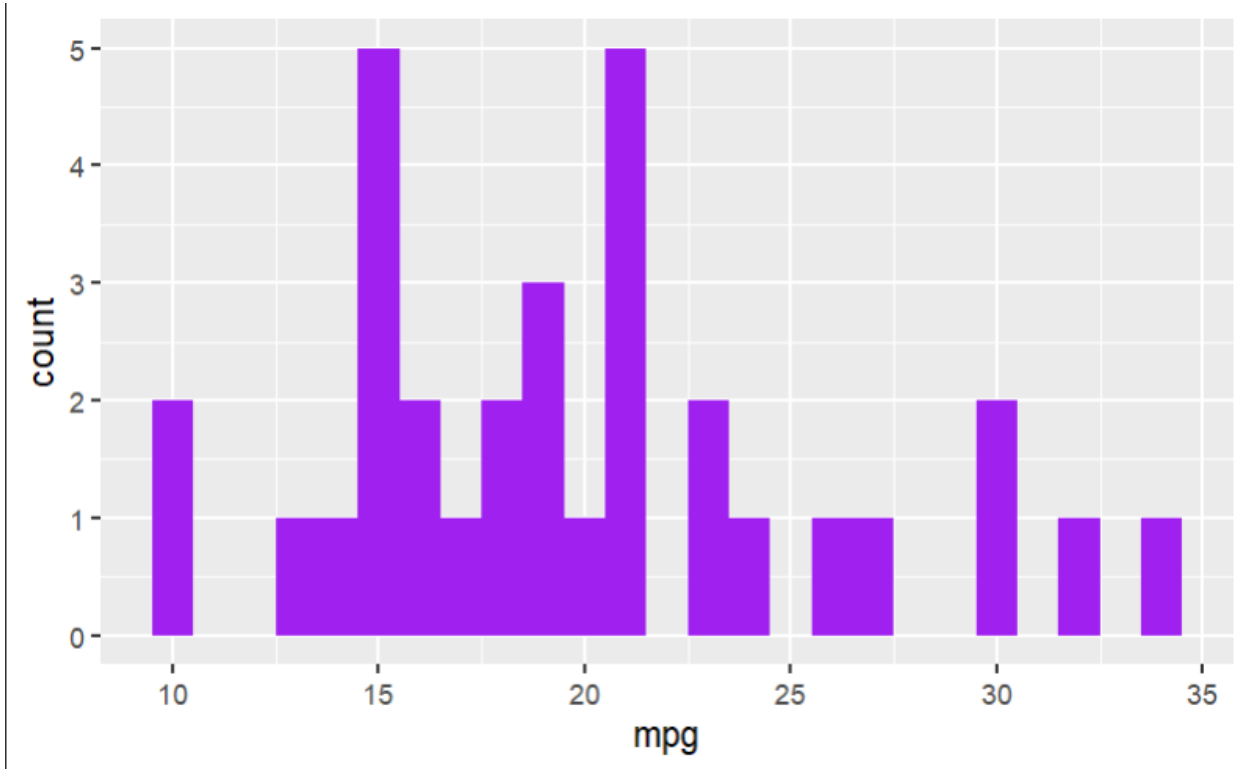
ggplot(freq_table, aes(x = factor(cyl), y = n, label = n)) +
  geom_text(size = 10, color = "darkblue") +
  labs(title = "Frequency Table Plot: Number of Cars by
Cylinder Type",
       x = "Number of Cylinders",
       y = "Count") +
  theme_minimal()
```



Numerical (univariate)

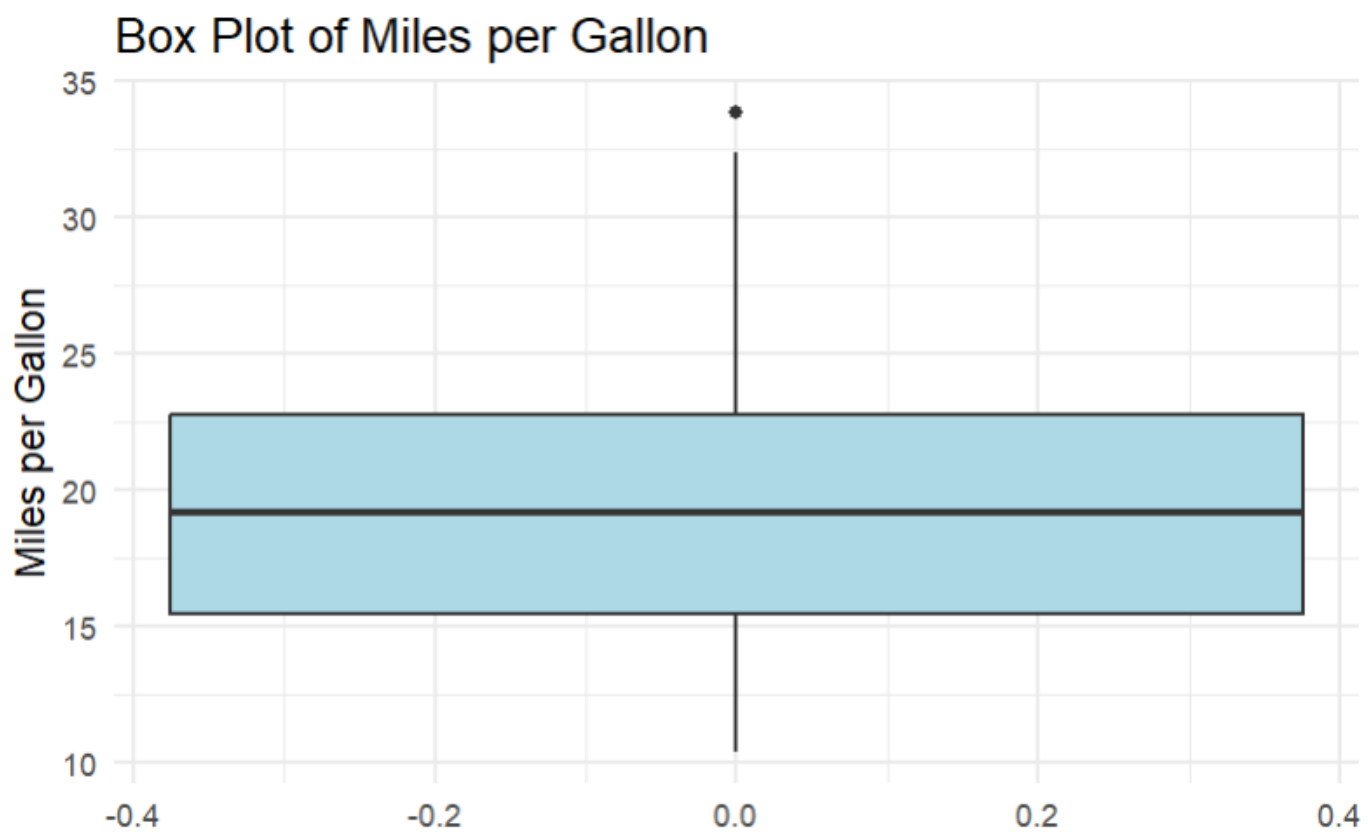
Histogram

```
ggplot(data = mtcars, aes(x = mpg)) +
  geom_histogram(binwidth = 1, fill = "purple")
```



Box Plot

```
ggplot(mtcars, aes(y = mpg)) +
  geom_boxplot(fill = "lightblue") +
  labs(title = "Box Plot of Miles per Gallon", y = "Miles
per Gallon") +
  theme_minimal()
```



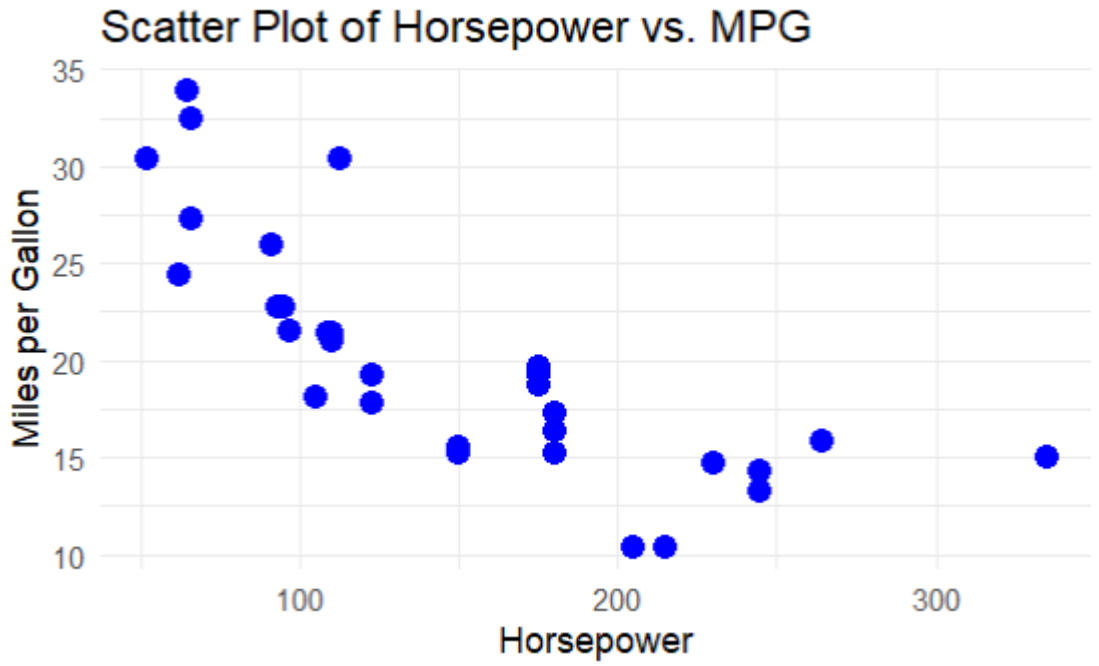
Numerical-Numerical Data

1. Scatter Plot: Shows the relationship between two numerical variables.
2. Line Plot: Visualizes trends between numerical variables (time series data).
3. Heatmap: Displays the density or count of values in a grid.

Scatter Plot

```r
library(ggplot2)

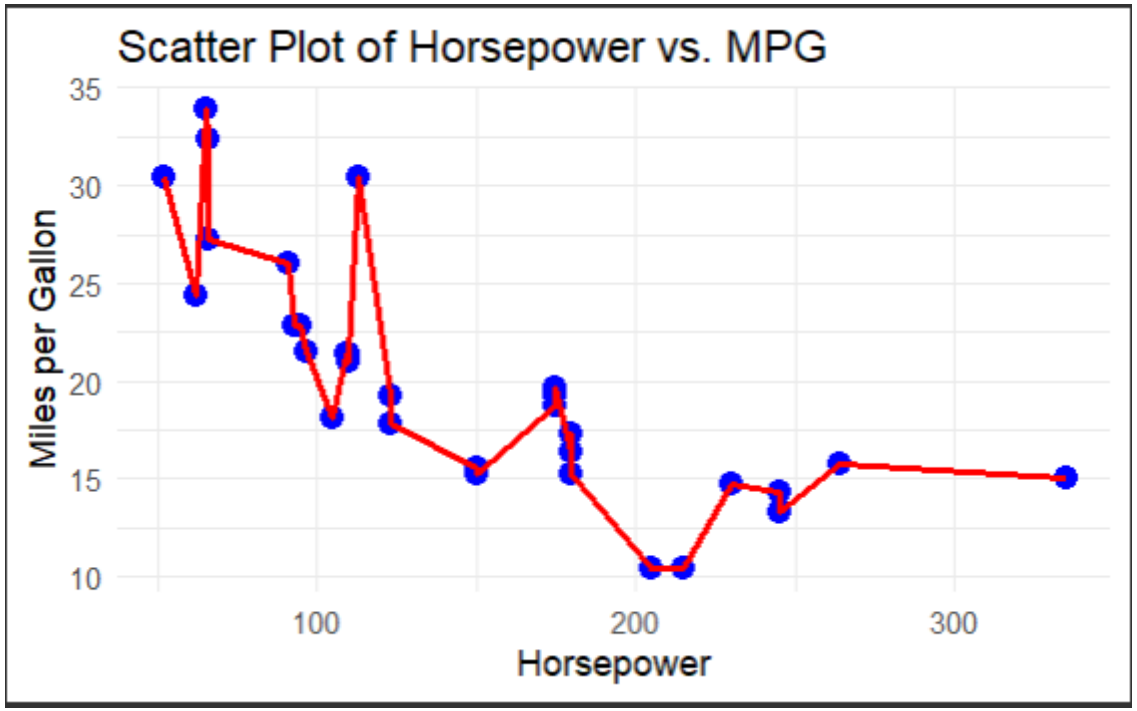
ggplot(mtcars, aes(x = hp, y = mpg)) +
  geom_point(color = "blue", size = 3) +
  labs(title = "Scatter Plot of Horsepower vs. MPG",
x = "Horsepower", y = "Miles per Gallon") +
  theme_minimal()
```



Line Plot

```r
library(ggplot2)

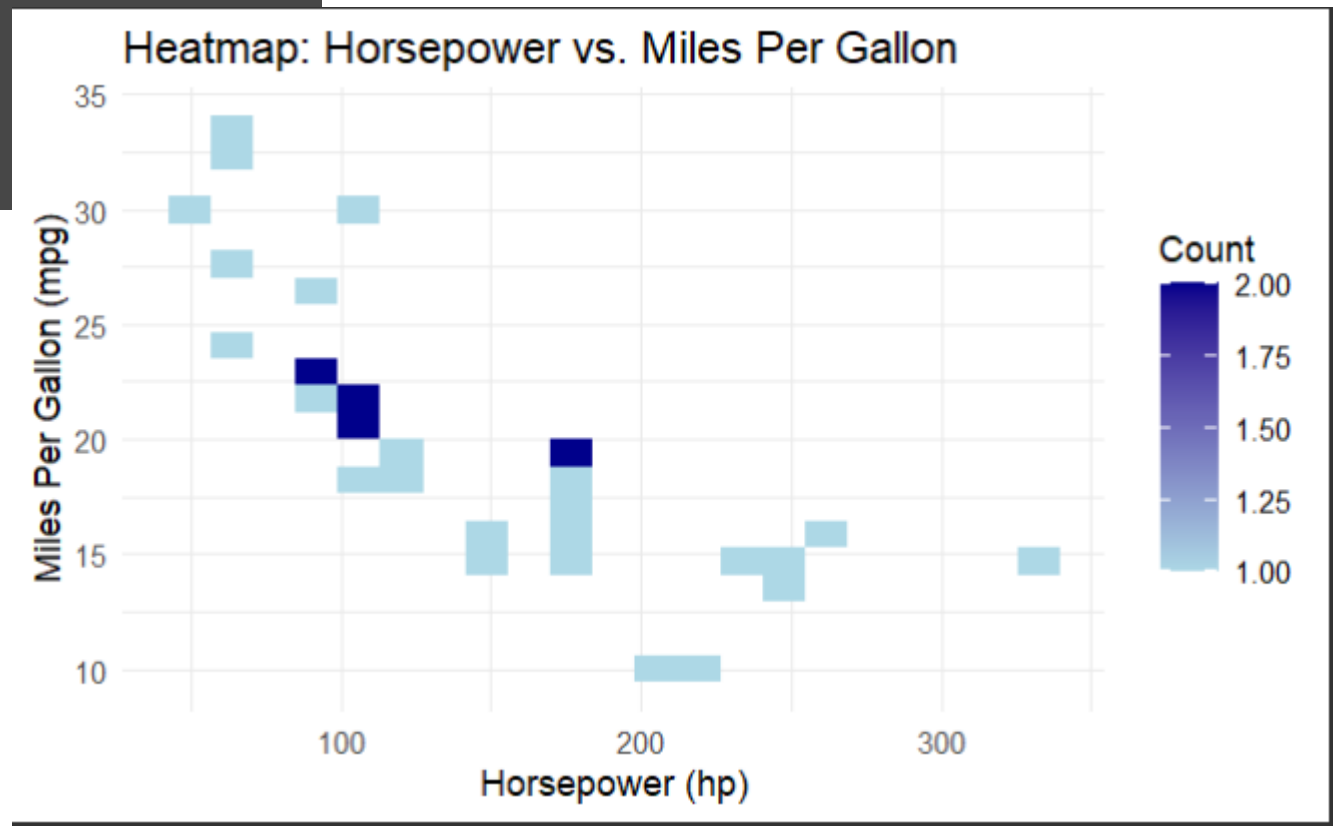
ggplot(mtcars, aes(x = hp, y = mpg)) +
  geom_point(color = "blue", size = 3)+
  geom_line(color = "red", size = 1) +
  labs(title = "Scatter Plot of Horsepower vs. MPG",
x = "Horsepower", y = "Miles per Gallon") +
  theme_minimal()
```



Heatmap

```r
library(ggplot2)

# Create a heatmap using geom_tile and bin2d to visualize the
density
ggplot(mtcars, aes(x = hp, y = mpg)) +
  geom_bin2d(bins = 20) +
  scale_fill_gradient(low = "lightblue", high = "darkblue") +
  labs(title = "Heatmap: Horsepower vs. Miles Per Gallon",
       x = "Horsepower (hp)",
       y = "Miles Per Gallon (mpg)",
       fill = "Count") +
  theme_minimal()
```

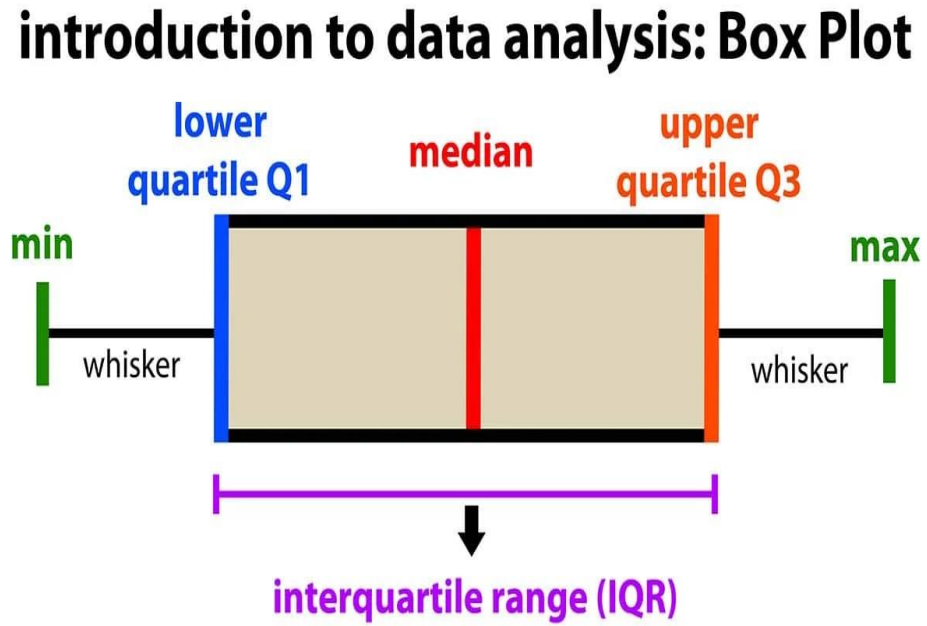


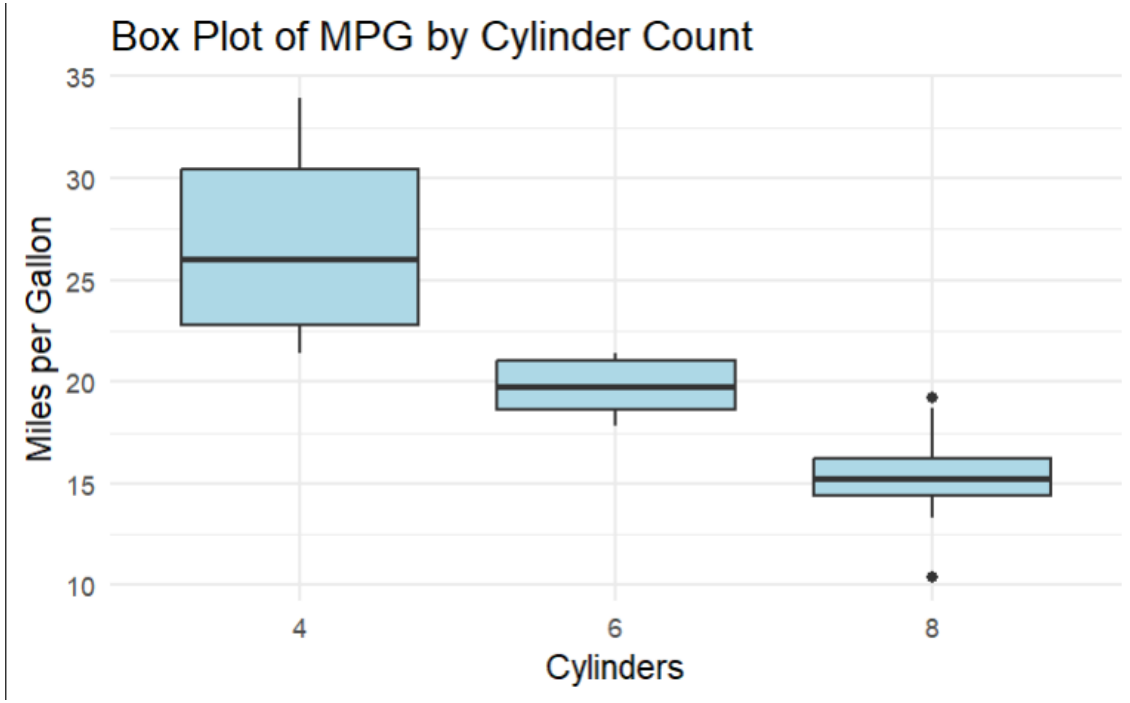
Categorical-Numerical Data

1. Box Plot: Shows the distribution of numerical data across categories.
2. Violin Plot: Combines the features of a box plot and a density plot.
3. Bar Plot: Displays summary statistics (mean, median) of numerical data for each category.
4. Histogram with Faceting
5. Density Plot

Box Plot:

```
# Box Plot of Miles per Gallon by Number of Cylinders
ggplot(mtcars, aes(x = factor(cyl), y = mpg)) +
  geom_boxplot(fill = "lightblue") +
  labs(title = "Box Plot of MPG by Cylinder Count", x =
"Cylinders", y = "Miles per Gallon") +
  theme_minimal()
```





Description about Box Plot

- **Minimum Score**

The lowest score, excluding outliers (shown at the end of the left whisker).

- **Lower Quartile**

Twenty-five percent of scores fall below the lower quartile value (also known as the first quartile).

- **Median**

The median marks the mid-point of the data and is shown by the line that divides the box into two parts (sometimes known as the second quartile). Half the scores are greater than or equal to this value, and half are less.

- **Upper Quartile**

Seventy-five percent of the scores fall below the upper quartile value (also known as the third quartile). Thus, 25% of data are above this value.

- **Maximum Score**

The highest score, excluding outliers (shown at the end of the right whisker).

- **Whiskers**

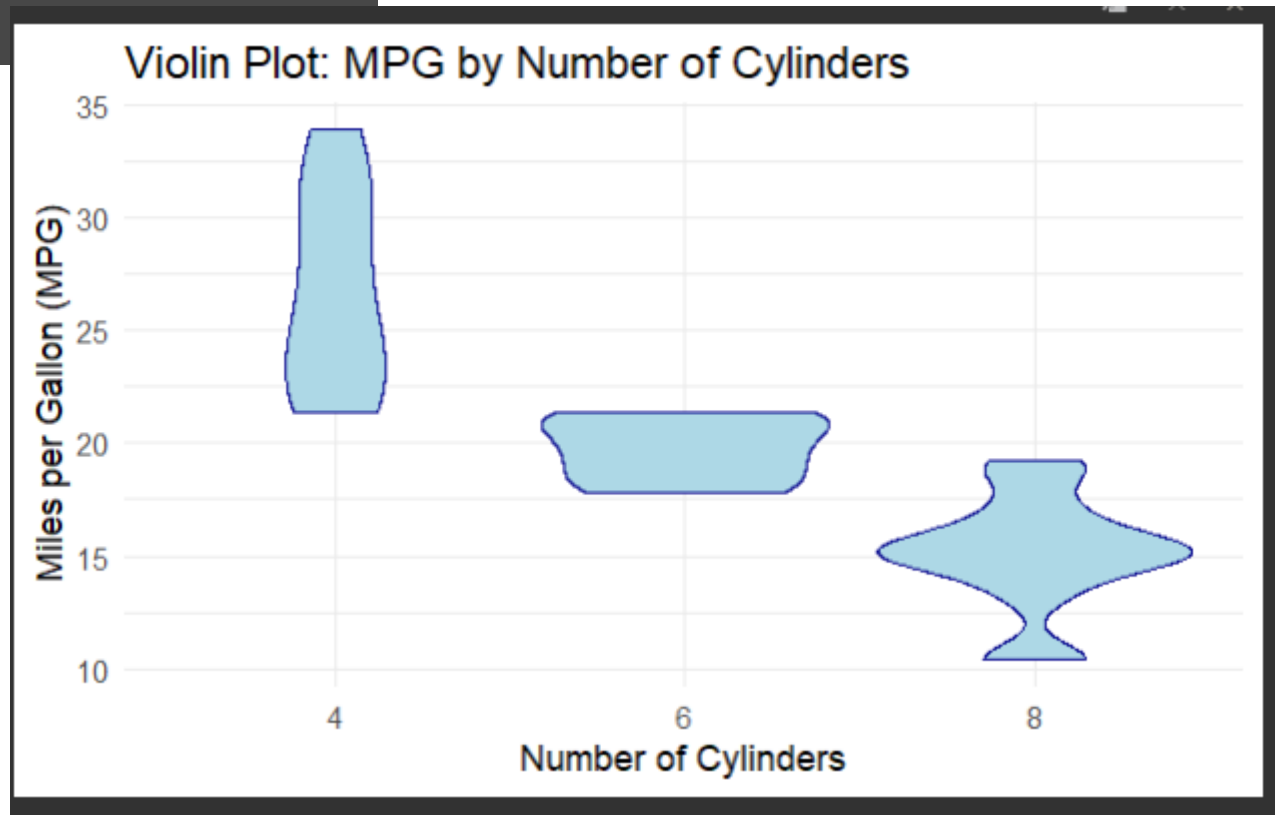
The upper and lower whiskers represent scores outside the middle 50% (i.e., the lower 25% of scores and the upper 25% of scores).

- **The Interquartile Range (or IQR)**

The box plot shows the middle 50% of scores (i.e., the range between the 25th and 75th percentile).

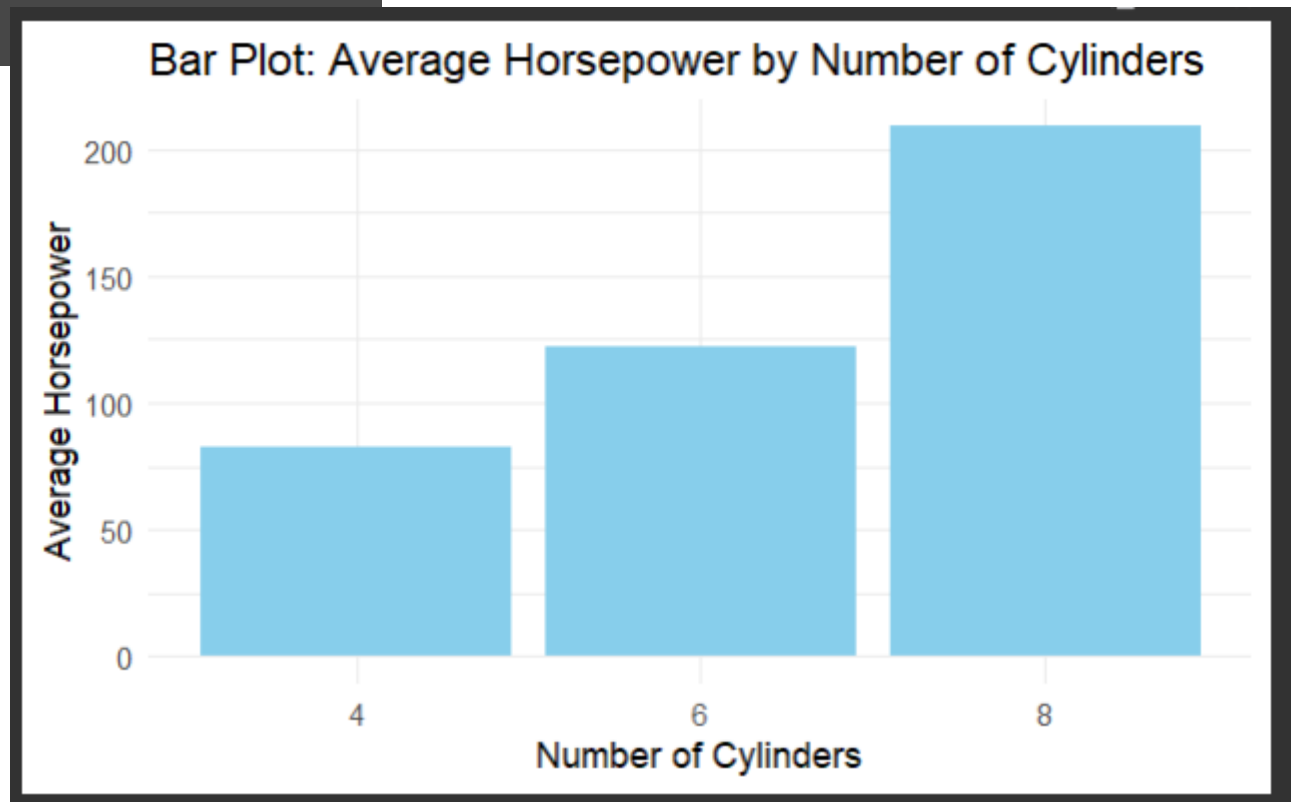
Violin Plot:

```r
# Create the violin plot
ggplot(mtcars, aes(x = factor(cyl), y = mpg)) +
  geom_violin(fill = "lightblue", color = "darkblue") +
  labs(title = "Violin Plot: MPG by Number of Cylinders",
       x = "Number of Cylinders",
       y = "Miles per Gallon (MPG)") +
  theme_minimal()
```



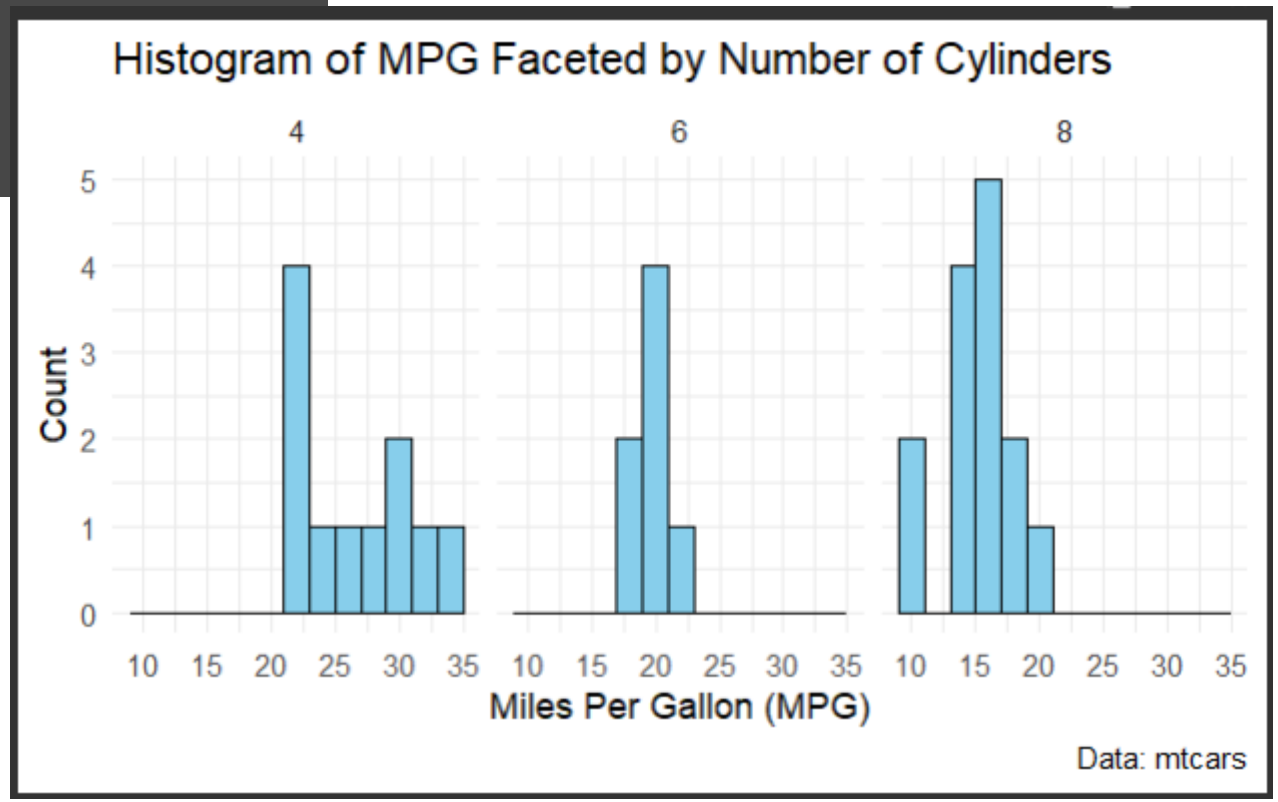
Bar Plot:

```
ggplot(data = mtcars, aes(x = factor(cyl), y = hp)) +
  stat_summary(fun = "mean", geom = "bar", fill =
"skyblue") +
  labs(title = "Bar Plot: Average Horsepower by Number of
Cylinders",
       x = "Number of Cylinders",
       y = "Average Horsepower") +
  theme_minimal()
```



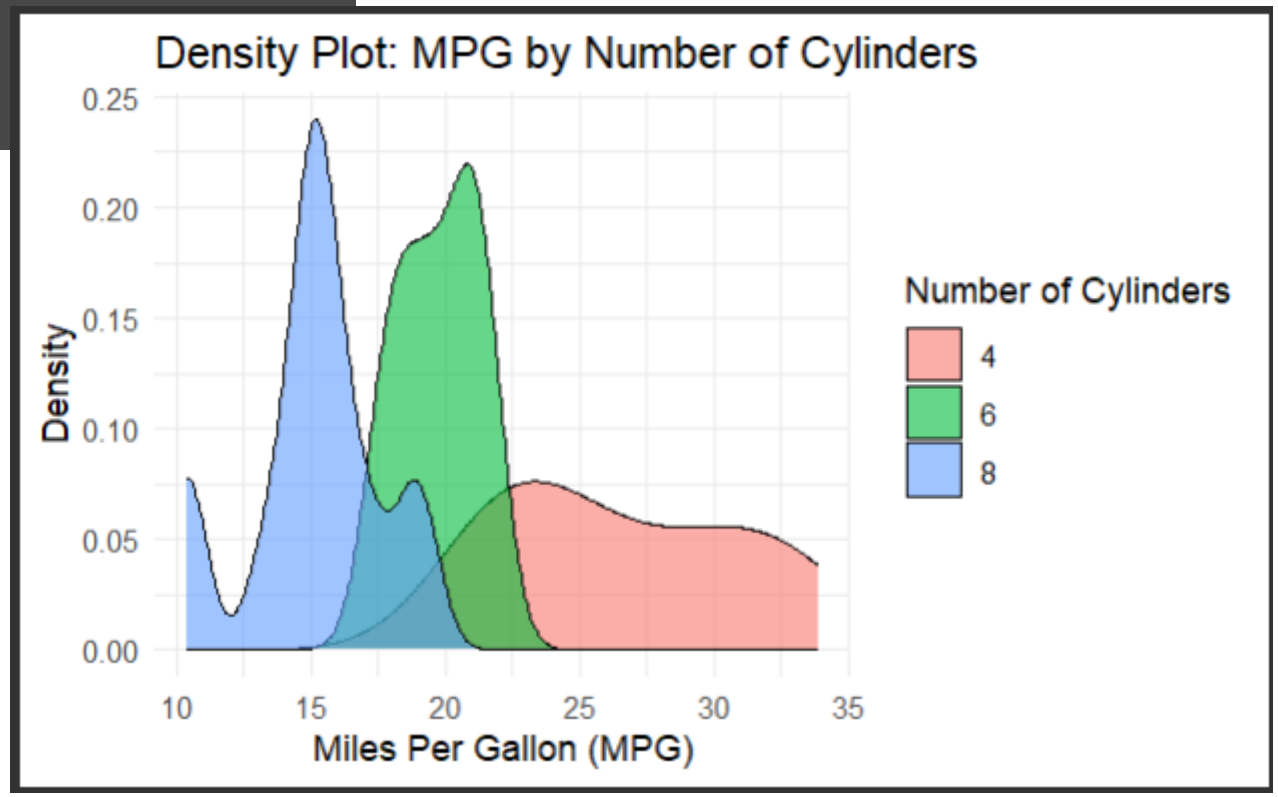
Histogram with Faceting

```r
ggplot(mtcars, aes(x = mpg)) +
  geom_histogram(binwidth = 2, fill = "skyblue", color =
"black") +
  facet_wrap(~ factor(cyl)) +
  labs(title = "Histogram of MPG Faceted by Number of
Cylinders",
       x = "Miles Per Gallon (MPG)",
       y = "Count",
       caption = "Data: mtcars") +
  theme_minimal()
```



Density Plot

```r
# Create a Density Plot of MPG grouped by Number of
Cylinders
ggplot(mtcars, aes(x = mpg, fill = factor(cyl))) +
  geom_density(alpha = 0.6) +
  labs(title = "Density Plot: MPG by Number of Cylinders",
       x = "Miles Per Gallon (MPG)",
       y = "Density",
       fill = "Number of Cylinders") +
  theme_minimal()
```



Categorical-Categorical Data

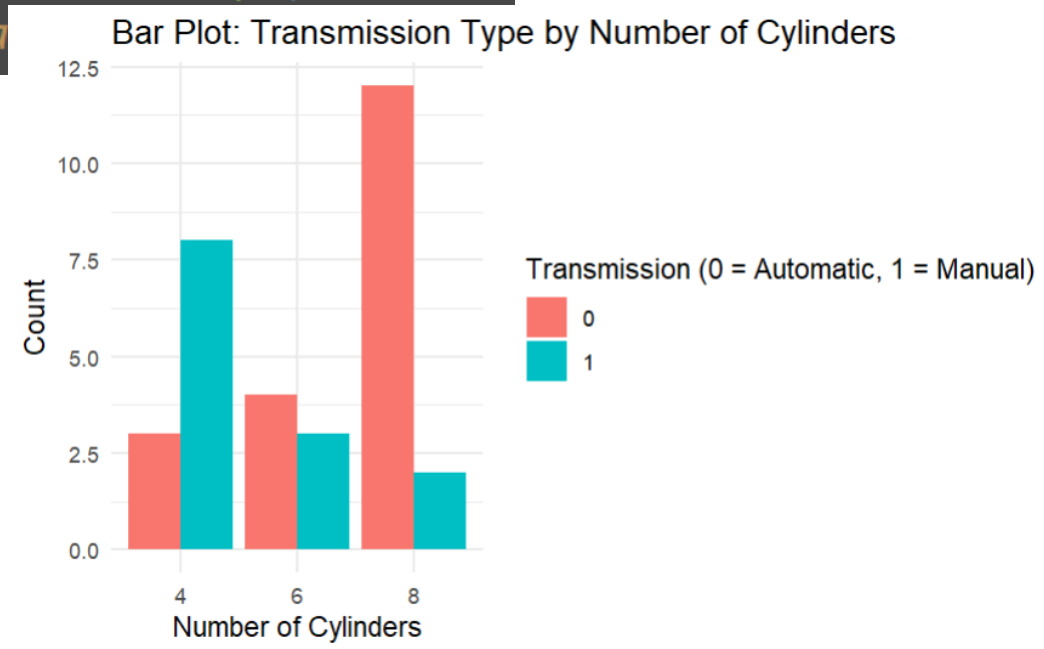
1. Mosaic Plot: Shows the relationship between two categorical variables using the area size.
2. Bar Plot: Compares the count or frequency of categories.
3. Stacked Bar Plot: Visualizes the proportion of subcategories within each primary category.

Mosaic Plot

```
ggplot(data = mtcars) +
  geom_mosaic(aes(weight = 1, x = product(factor(cyl)),
fill = factor(am))) +
  labs(title = "Mosaic Plot: Number of Cylinders vs.
Transmission Type",
       x = "Number of Cylinders",
       y = "Proportion",
       fill = "Transmission (0 = Automatic, 1 = Manual)")
+
  theme_minimal()
```

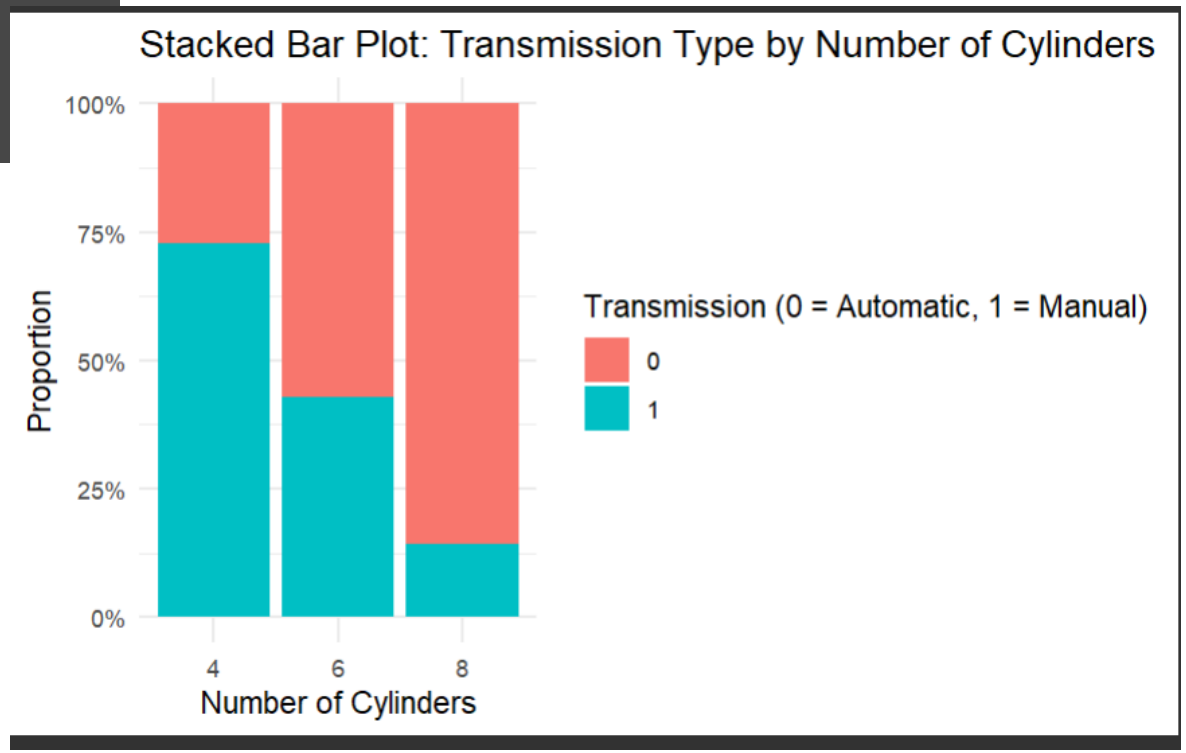
Bar Plot:

```r
# Create a side-by-side bar plot to show the count of cars by
number of cylinders and transmission type
ggplot(mtcars, aes(x = factor(cyl), fill = factor(am))) +
  geom_bar(position = "dodge") +  # Use "dodge" for side-by-side
bars
  labs(title = "Bar Plot: Transmission Type by Number of
Cylinders",
       x = "Number of Cylinders",
       y = "Count",
       fill = "Transmission (0 = Automatic, 1 = Manual)") +
  theme_minimal()  # Use a minimal theme for a
```



Stacked Bar Plot:

```r
# Create a stacked bar plot to show the count of cars by number of cylinders and transmission type
ggplot(mtcars, aes(x = factor(cyl), fill = factor(am))) +
  geom_bar(position = "fill") +  # Use "fill" for stacked percentage
  labs(title = "Stacked Bar Plot: Transmission Type by Number of Cylinders",
       x = "Number of Cylinders",
       y = "Proportion",
       fill = "Transmission (0 = Automatic, 1 = Manual)") +
  scale_y_continuous(labels = scales::percent) +  # Format y-axis as percentages
  theme_minimal()  # Use a minimal theme for a clean look
```



THANK YOU